ConseqaX: An Evidence Protocol for Recovering Ambiguous Agent Actions

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***Abstract*—A timed-out tool call can leave an agent uncertain whether an operation failed or will commit later. ConseqaX binds temporal contracts, readback witnesses separate from acknowledgements, and distinct state and action-origin conclusions to an application recovery policy. We evaluate five deterministic policies under shared provider guarantees. Study A retains all 15,360 SQLite worlds across 24 authored cases in 12 purposive families; its 128 seeds added no recorded-metric variation within a case–policy group. Four readback policies achieved safe recovery in 14 cases, versus eight for response-driven retry. The package matched a strong reference with separate evidence arithmetic and shared application transitions. Its original-contract state reports were wrong in 0/2,304 definite outputs, while harmful effects remained possible. Polling doubled reads without improving final claims under these schedules. A separately frozen 960-world follow-up varies 64 availability profiles across three domains: polling resolves 27 profiles versus one for a single final read, at 741 versus 192 total reads per policy. The reference shares this gain; physical outcomes remain identical. The contribution is an inspectable evidence protocol and conditional coverage–cost evaluation. A postcollection packaging deviation in Study A is disclosed. These internal experiments do not establish autonomous-agent superiority or transfer to independently operated services.**

***Index Terms*—agent systems, outcome verification, recovery, idempotency, evidence provenance, uncertainty**

I. Introduction

An agent requests an inventory allocation, ticket, or published artifact. The provider accepts the operation, but the acknowledgement times out. An immediate retry may create a second record; stopping may leave the task unfinished. A separate read can reveal an existing effect, yet even a fresh empty snapshot cannot establish that the first request will never commit later. A reliable recovery decision depends on the operation's identity, provider guarantees, temporal target, and available evidence, rather than the response alone.

Three distinctions matter. First, a response describes a tool interaction, whereas a readback supplies evidence about external state. Second, global task satisfaction differs from an effect being linked to this agent's operation. Another actor can complete the task; missing actor metadata need not imply an external actor. Third, a hindsight-correct assertion can have been unsupported when issued. A controller that guesses the eventual outcome correctly has not thereby justified a retry or a completion claim.

These are established systems and verification concerns [1], [2], [3], [4], [5], [6]. ConseqaX makes them explicit in a reusable evidence protocol [7]. Its typed records bind contracts, attempts, responses, observations, and verification receipts. Admitted readback can support satisfied, violated, or unknown state, while origin remains a separate conclusion. The protocol does not authenticate a provider or manufacture request finality. Applications supply source semantics and the policy connecting receipts to actions.

This paper centers that policy boundary. We compare response-driven same-key retry, state-only readback, a separately implemented provenance-aware reference, a package-backed single-final-read ablation, and package-backed bounded polling. Controller actions change subsequent SQLite effects and observations. The experiment asks whether the declared evidence policy changes safe recovery, final-report correctness, or definite-report coverage at a measured read/wait cost, and when ordinary provider idempotency removes the apparent advantage.

The research questions are falsifiable. **Q1:** How does polling change safe recovery relative to response-driven retry, together with own harmful effects and recovery failures? **Q2:** Does the package-backed implementation agree with a same-information reference using separate evidence arithmetic? **Q3:** What additional origin-report coverage and correctness appear beyond state-only reporting? **Q4:** Does continued polling change final-report correctness and coverage relative to one final read at its additional resource cost? **Q5:** What abstention and definite-error tradeoffs remain under incomplete public histories? Equal or worse recovery, incorrect conclusions, unnecessary reads, or parity under shared provider guarantees can limit the proposed benefit.

The contribution is a specified application protocol, a bounded recovery policy, and a paired closed-loop evaluation of these questions. Separately implemented reference parity would support observed conformance, without demonstrating superiority. Because state-only and provenance-aware variants share action transitions, their comparison isolates reporting rather than a provenance-caused recovery benefit. The 12 purposive families provide challenge-suite evidence; repeated seeds do not establish a population of independently sampled tasks.

II. Related Work and Positioning

Idempotence and eventual consistency are longstanding systems concerns [1], [2]. Verified Tool Calls separates acknowledgements from independently checked effects and admits unknown postconditions [3]. Callability Is Not Operability examines controlled interface ambiguity and terminal agent

claims [4]. Agent-tool anomaly work distinguishes runtime observations from external events [5]. LIMBO grades committed-effect histories and studies late commits [6]. Consequently, response/effect separation, effect ledgers, and the danger of ambiguous retries are foundations and comparison requirements here, rather than priority claims.

PROV-DM represents entities, activities, agents, and responsibility [8]. ConseqaX is neither a new general provenance model nor a PROV conformance implementation. Its narrower boundary links operation identity, temporal contracts, individual observation witnesses, and explicit unknown verdicts. A source-supported origin link is not proof that the action was counterfactually necessary.

Causal Agent Replay (CAR) performs same-policy interventions with contrastive suffix and permutation Shapley attribution [9]. Policy-Conditioned Counterfactual Credit studies verified outcomes under declared interventions and continuation policies [10]; CausalLoss-Fin jointly intervenes on decisions and infrastructure messages [11]. Those questions differ from whether a live recovery action is justified by its available evidence. Our earlier replay adapter uses existing estimators. The present experiment requires neither counterfactual writes nor a new attribution method.

AppWorld, ToolSandbox, and τ-bench already evaluate stateful agent interaction and world outcomes [12], [13], [14]. A maintained retail/airline corpus and native DB evaluator also support our earlier model-free transfer experiment [15], [16]. Replaying reference actions or executing deterministic recovery logic does not supply an autonomous benchmark score. The next external test must retain native task reward, actual agent decisions, matched classical recovery controls, and a separately checked side-effect ledger.

The present contribution is therefore conditional: an executable evidence contract and an auditable comparison of policy tradeoffs under declared capabilities. Its value must come from the recovery behavior, final-report correctness, coverage, and cost actually measured, rather than from attributing established distributed-systems guarantees to a framework.

III. Evidence and Recovery Protocol

A. Objects and temporal targets

A logical action has stable identity and payload; each dispatch has a distinct attempt identity. Captured records include the request, provider response, reported assessment, separately captured observation, and resulting verification receipt. Reported assessments are outputs, not measurements of hidden reasoning. An observation states its source, scope, completeness, availability, observation time, watermark, and effect history. The application declares which source and clock are authoritative.

Let registration be t_0, the original deadline D, stability duration W, and original horizon $H = D + W$. The original contract remains fixed. A separate recovery deadline $R \geq H$ and safety horizon $S \geq R$ define later recovery in this study: its matching candidates must commit by R and persist through S, and its count is checked throughout $[R, S]$. A later completion may satisfy recovery while violating the original SLA. Neither a new read nor an explanatory receipt moves D.

For the exact-count record contract, a candidate matches the expected business fields, commits within $[t_0, D]$, and persists through H. The declared count must hold, including the maximum simultaneous matching count over $[D, H]$. Historical removals and transient extras therefore matter; a current-row snapshot alone is insufficient. Simultaneous commits and removals are evaluated together, with boundary conventions fixed before collection.

The verifier concludes satisfied, violated, or unknown. A complete admitted observation must cover the registered scope and history through H. Stale watermarks, inadequate scope, unavailable reads, or malformed history retain unknown. Source identity, completeness, timestamps, and provenance are trusted adapter declarations. Schema checks and hashes establish structure and retained bytes, not source authenticity or transactional guarantees.

B. State, origin, and support

Global state satisfaction does not require this agent to have produced the effect. Action-linked satisfaction additionally requires qualifying target evidence linked to the logical operation. A known global violation makes this linked contract false even if an own matching record exists: a second effect can invalidate an exactly-one task. Mere witness presence cannot replace the whole contract.

A matching action identity with absent or matching actor metadata can support linkage under the admitted provenance law. Explicitly different identities exclude this operation; missing or conflicting linkage can remain unknown. Provenance does not turn an externally satisfied state into a failure. Each observation retains its own witnesses. A later record sharing an effect ID cannot inherit an earlier observation's origin link while silently changing payload, actor, or removal history.

Hindsight correctness and evidence support are distinct questions. The present evaluator scores final state and action-linkage reports against full-ledger truth, with separate definite-report and abstention counts. It does not independently label support at issuance or score safe-to-retry assertions. Retained public histories permit automated checks against the declared admission rules; a hindsight-correct guess alone does not establish support. An origin abstention does not erase a definite state conclusion. Conditional error among definite assessable reports and correct definite yield over all planned worlds answer different questions.

C. Why historical absence cannot resolve a pending attempt

Conditional retry boundary. Suppose an operation times out without atomic deduplication, terminal resolution, completed cancellation, or an enforceable in-flight bound. Let $L < R$ be the last useful retry time. Two provider worlds give identical admissible histories through L: the initial attempt never commits in one and commits after R but by S in the other. A retry by L achieves recovery by R; a later retry

cannot. An extra effect is harmful even if compensated, and no alternative operation achieves the target.

A deterministic policy takes the same decision in both histories. Retrying creates a duplicate in the late-commit world; declining misses recovery in the never-commit world. For a randomized policy, couple its randomness and let p be its useful-retry probability. Duplicate risk in the second world is at least p and missed-recovery risk in the first at least $1 - p$. Their sum is at least one for this constructed pair. This elementary indistinguishability argument states a known systems boundary [6], rather than a new impossibility or a measured prevalence. Provider guarantees can break its assumptions.

D. Application safety gate and bounded reads

Every policy preserves the operation key and payload. When a public native idempotency capability is admitted, the provider atomically reserves pending and completed work for that key. All readback policies may use the same-key exception to retry within the common cap before D. The response-driven arm also benefits from the identical provider semantics; it is the ordinary bounded idempotent-retry comparator in that stratum.

Without that capability, the readback policies permit another dispatch only when every previous attempt has an admitted terminal-no-effect guarantee, fresh complete trusted readback establishes no relevant effect through the decision time, budget remains, and time is strictly before D. A newer terminal response does not resolve an older ambiguous request. An empty watermark proves only past absence; it supplies no terminal guarantee. Resolution of the agent's attempts also cannot prevent another actor's future write. Global race freedom would require additional provider semantics.

The package's generic retry advice is recorded, but this application gate authorizes dispatches. Provisional predeadline verification has its own target ending at the current time; it never replaces the fixed D/H receipt. Bounded polling uses externally scheduled callbacks rather than the public blocking polling helper, allowing all arms to share admitted opportunities. At exhaustion, insufficient evidence remains an abstention. No policy dispatches after D; recovery at R is a distinct evaluator outcome, not an extension of the original dispatch window.

IV. Frozen Closed-Loop Study

A. Corpus, separation, and execution

The schedule crosses 24 new cases, 128 timing seeds, and five policies: 15,360 planned worlds. Cases span 12 purposive families in three nonpayment SQLite domains: inventory, tickets, and artifacts. Related cases share mechanisms; identifiers and seeds do not create new families. Separate AI agents authored controllers and cases. The controller author never inspected the case file, and the lead integrator did not manually inspect case values before freeze. This is procedural separation within an AI-assisted workflow; the tasks are not externally authored or randomly sampled from a task population.

Each world uses a fresh database. The frozen exogenous event schedule, timing seed, provider law, source behavior, budgets, and read opportunities are paired across policies. Actions create branch-specific effects, so later observation contents can legitimately diverge. Matching means equal capabilities and opportunities, not forcing one policy to consume another's realized evidence. Readback is a separate application method on the same SQLite connection and ledger. It supplies a signal separate from the acknowledgement, rather than an independent connection, source, or trust domain.

The public allowlist includes request/contract, identities, response history, provider capability and terminal receipts, and scoped readback history. Controllers receive no case/family/fault labels, hidden future events, evaluator truth, or another policy's reports. They are deterministic application state machines. The run requires no LLM, user simulator, remote provider, or paid API call and supplies no autonomous-agent benchmark reward.

B. Five policies and the strong comparison

The single-read arm is not forced to spend its final query before H. It may perform a predeadline gate query when terminal guarantees make retry relevant; its first query at or after H is its only final query. Polling can use later opportunities to resolve lag, incompleteness, or unavailability. Equal maximum budgets need not yield equal spend; early stopping and skipped queries are policy outcomes.

The reference imports no package verifier. The controller author implemented its evidence arithmetic separately, while sharing the declared application transition logic with the polling comparison. Thus exact agreement is a conformance result, not a contest between independently designed recovery algorithms or external implementations. State-only and provenance-aware arms likewise do not test a provenance-caused action benefit when transitions coincide. They test origin-report coverage and correctness under the admitted provenance law.

C. Endpoints, costs, and finite-suite analysis

The primary all-slot endpoint is **operational safe recovery**: the global recovery payload/count contract holds by R and persists through S, with neither an own duplicate nor an own unwanted effect committed through S. An own duplicate means more cumulative own committed effects than the allowed logical-operation count, including effects later removed. An own unwanted effect mismatches the expected business fields or commits after R. These predicates do not certify the absence of every harmful external activity; another actor's effects can still violate the global recovery count. Every scheduled cell remains in the operational denominator, and failed or unstarted execution cannot earn success. Failed cells' semantic state, origin, and harm outputs remain unavailable rather than receiving fabricated violation labels; an incomplete collection cannot yield a complete audited result. Own duplicate and unwanted events are mandatory companions, including effects

TABLE I

FROZEN POLICIES, EVIDENCE INTERPRETATION, AND RECOVERY/READ BEHAVIOR. NATIVE IDEMPOTENCY AND TERMINAL-RESOLUTION CAPABILITIES ARE SHARED; INDEPENDENT REFERENCE ARITHMETIC DOES NOT IMPLY AN INDEPENDENTLY DESIGNED APPLICATION POLICY.

Policy	Evidence interpretation	Recovery and reads
`response_retry`	ACK/error proxies for state and action completion; unresolved timeout unknown.	Retry ambiguous timeouts with the same key within the cap and original deadline.
`readback_state`	Trusted complete history for global state; abstain on provenance when state holds.	Shared gate and bounded scheduled reads.
`reference`	Independent standard-library count/time and provenance arithmetic.	Same application state machine and read opportunities as polling.
`consequax_one_read`	Public 2.1.1 verification receipts plus explicit linkage uncertainty.	Gate-relevant predeadline queries; one final horizon-eligible query.
`consequax_poll`	Same public verification and linkage rule.	Shared gate with continued scheduled reads until known or exhausted.

subsequently removed. If safety is unauditable, all-slot harm lower/upper bounds must expose that uncertainty.

Retained truth records distinguish original D/H satisfaction, operation-linked satisfaction, and controller-created effects. The summary scores the final state and action-linked reports for hindsight correctness, false success, definite coverage, and abstention. Conditional error is wrong definite outputs divided by definite assessable outputs; correct definite yield uses all planned worlds. A zero-claim error rate is undefined. Semantic rates and resource means use observed completed cells with unavailable counts shown; the primary operational score and claim yields retain planned denominators. External completion does not establish successful execution by this controller.

For family f and policy m, let y_{fm} be mean operational safe recovery over every scheduled case/seed cell in that family. The predeclared primary contrast is

$$\Delta = \frac{1}{12}\sum_{f=1}^{12}(y_{f,\text{poll}} - y_{f,\text{response}}). \tag{1}$$

The full 12 paired differences accompany the equal-family mean and all-slot cell totals. Secondary views compare polling with state-only and one-read; reference agreement is implementation checking. Capability, domain, and fault strata are descriptive. No population p-values, task-population confidence intervals, equivalence, or noninferiority claims follow from these purposive families. Timing-seed replication characterizes this fixed suite, not new independent tasks. Any seed-law uncertainty must identify its own sampling assumptions and cannot supply task-population inference.

The analysis retains actual dispatches, readbacks, wait decisions, and logical decision time under shared ceilings, separating evaluator audit work. Accounting includes unsuccessful requests and timeout dispatches. Provider terminal guarantees are response fields; separate resolution, cancellation, and refused-request cost endpoints are not implemented. A zero-success cost-per-success ratio is undefined. Host runtime is separate from simulated logical time; one local collection does not establish intrinsic production overhead, wall latency, or service charges.

D. Freeze and first-run integrity

The pinned 2.1.1 wheel runtime is bound to study source `22e88498cff0e2afd1549803fb4d9645902e74b9` and protocol commit `3d3dd4ca663d3e16c7cb1ead4c59c724af9bedd7`. Protocol SHA-256 is `b5c6a06bf1d758678e4d9890aa66872d4036634e2a4a4fd89626bd710bbc5aaa`. The artifact binds corpus, schedule, controllers, provider/evaluator law, analysis, Python executable identity, and installed framework source/runtime before measured execution. It retains first-run cells, failures, stop/abstain decisions, and deviations. Corrections after exposure require a separate version; deterministic re-execution cannot replace the original denominator or create new independent trials. Hashes identify bytes rather than authenticating the source's semantic claims.

A packaging deviation is retained explicitly. All scheduled trials, database snapshots, journal rows, and the summary were present when the session resumed, but the original runner's terminal manifest and receipt were absent. Its exit status and termination cause are unknown. A separate post-collection finalizer preserved every existing byte, rechecked source/runtime/claims, and created only the missing packaging files. It did not resume or rerun worlds or regenerate the summary. The completion timestamp denotes packaging time. The unchanged frozen auditor subsequently checks all retained semantics; neither this audit nor retained claims prove absence of off-record execution or earlier tampering.

V. RECOVERY RESULTS

A. Retained collection and effective diversity

All 15,360 scheduled worlds are retained, with 3,072 completed cells per policy and no execution failures. The unchanged auditor verified every SQLite snapshot and recomputed the endpoint summary; a portable artifact recheck agreed. The packaging deviation in Section IV-D remains part of this result. Across all 120 case–policy groups, none of the recorded endpoint or resource metrics varied over the 128 seeds. Some exogenous timestamps varied, but the repetitions added no endpoint diversity. The substantive challenge set therefore remains 24 authored cases in 12 purposive families, rather than 15,360 independent tasks.

B. Recovery, harms, and the strong comparison

Every readback policy achieved operational safe recovery in 1,792/3,072 worlds (14/24 cases), compared with 1,024/3,072 (8/24) for response-driven retry. The predeclared equal-family primary difference was +0.25. Figure 2 shows all 12 family contrasts: three were +1, and nine were zero. Differences against the reference, state-only, and one-final-read policies were zero in every family.

The three positive families have specific explanations. In weather-supply and translation-routing cases, response-driven timeout retries created an extra non-idempotent effect; the readback policies dispatched once. In mobile-lab cases, the first response explicitly terminated without an effect; readback policies verified fresh absence and retried, while the response baseline's declared error rule stopped. This third gain concerns that rule, rather than superior deduplication. The equally informed classical reference achieved the same recovery as the package.

Residual harm is material. In the same 256 archive-rights worlds, one initial provider dispatch created two wrong-payload effects. These are both duplicate and unwanted events, not 512 distinct harmed worlds. Detection did not cancel or repair the effects. Environmental-release cases supply a different failure: an external actor satisfies the original contract, then a delayed own effect breaks the later exact-count target without an own duplicate or unwanted flag. All 256 response worlds with original satisfaction but failed recovery belong to this family. Equal aggregate original/recovery counts for the readback policies conceal a 256-world later-recovery gain and a different 256-world later loss. One target cannot substitute for the other.

All 3,072 reference–polling pairs agreed on final state/linkage reports, endpoint vectors, controller action sequences, and full retained truth. This is observed implementation conformance under the shared policy, not superiority over the strong reference. State-only and polling also had identical actions and physical histories; 1,024 pairs differed in the action-linked report. Provenance expanded reporting here, without a recovery-action benefit.

C. Reporting and the cost of additional polls

The response policy made 768 false-success and 256 false-negative state reports. The readback policies made no wrong definite state report, but abstained in 768/3,072 worlds. Provenance-aware policies retained 1,024/3,072 linked abstentions; state-only retained 2,048/3,072. Correct negative reports contribute to these yields. Zero wrong original-contract reports did not imply zero harms or guaranteed safe recovery.

Polling and one-final-read agreed on both final claims, dispatches, effect histories, and every non-polling-cost endpoint in all 3,072 pairs. Polling nevertheless consumed 7,680 reads versus 3,840, with mean logical decision time 220 versus 200 ms. These are simulated decision times, not measured production latency. The schedule explains the negative finding: no case with insufficient evidence at H became admissibly known at $H + 40$ or $H + 80$. Transient limitations either cleared before the first final read or persisted beyond the last opportunity. Thus Study A shows no incremental polling utility under its schedules; it does not test the general benefit of recoverable lag. The separately frozen follow-up below addresses that specific coverage gap.

D. Study B: controlled evidence availability

Study A's coverage gap motivated a separate follow-up after its outcomes were known. We retained the byte-identical world, controllers, and evaluator, then froze a new complete factorial grid before collecting its own data. Source commit `d805b3e3560c05ee0a7efd02a80fc18503c6ee05`, protocol commit `b937920469285cdd971783fa6cb33216a7e4da13`, and protocol SHA-256 `e874c6781aa0dbc8ef682ea0a8d37c25d73cc861dada05fd0379116d4a7f6584` bind that design. This is a disclosed, post hoc motivated sensitivity experiment, rather than a new held-out corpus or replacement for Study A.

The 64 profiles cross readback lag $\{0, 40, 80, 95\}$ ms with independently varied unavailable-until and incomplete-until times $\{0, 170, 210, 250\}$ ms. Each profile repeats in the three domains at seed zero: 192 cases and 960 scheduled policy worlds. Here $D = 100$, $H = 150$, $R = 250$, and $S = 330$ ms; reads remain at $50, 150, 190, 230$ ms. Every initial dispatch acknowledges at 2 ms and creates one correct effect at 10 ms. All arms share native idempotency. There are no external writes, removals, or wrong effects. This intentionally uniform physical-success design isolates evidence-window coverage and cost. Profiles are controlled mechanisms, and the three domains repeat them; neither 192 cases nor 960 worlds represent independent task samples.

All 960 worlds completed without failure. The separately implemented auditor verified 1,925 saved artifacts, queried all 960 SQLite snapshots, and agreed with the summary. Unlike Study A, the original collector produced its terminal manifest and receipt normally. All five policies dispatched once, achieved original and recovery satisfaction in 192/192 worlds, and had zero duplicate or unwanted events. Their physical effect histories agreed in every five-arm case. Thus the primary safe-recovery contrast is zero.

The one-final-read policy resolved one of 64 profiles per domain; polling resolved 27, matching the prospectively specified timing calculation. The paired correct-state-yield increase is $78/192 = 26/64 = 0.40625$, shared by the reference and state-only readback. All 192 reference–polling pairs agreed on reports, endpoint vectors, dispatch decisions, and physical records. State-only differed from polling in 81 linked reports while preserving actions and physical outcomes. Polling used 741 reads versus 192 for one-final-read, or 3.859375 versus one per world; mean logical decision time increased by 74.375 ms. These are charged simulated reads and decision times, not production latency.

The full factor grid in Fig. 3 shows the boundary: late opportunities help only when availability and completeness recover and the watermark reaches H. Lag 95 ms and either window persisting through 250 ms remain unresolved. The ACK-driven response policy reports all 192 successes correctly in hindsight

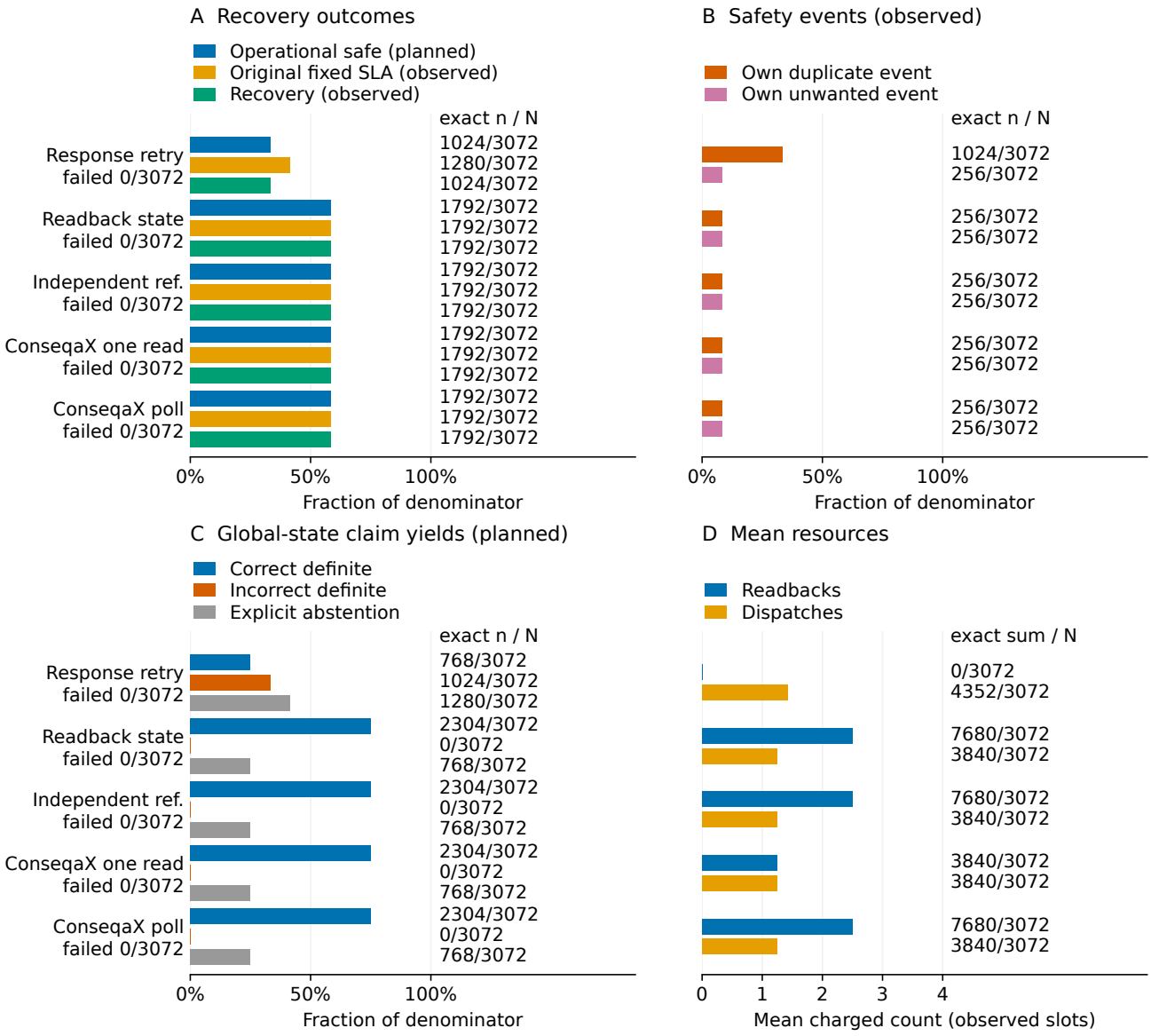


Fig. 1. Study A: all five policies, exact denominators, own harmful events, global-state reporting, and read/dispatch costs. All recorded metrics were constant across seeds within each of the 24 authored cases. Reference independence concerns evidence arithmetic; application transitions and controller authorship are shared. Reporting correctness is distinct from physical recovery and evidence support.

TABLE II

STUDY A: RECOVERY AND RESOURCES. EACH POLICY HAS 3,072 PLANNED AND COMPLETED WORLDS. ORIGINAL D/H SATISFACTION AND RECOVERY R/S ARE DISTINCT TARGETS. OWN DUPLICATE AND UNWANTED EVENTS CAN OVERLAP. MEANS ARE EXACT CHARGED COUNTS, NOT WALL LATENCY.

Policy	Safe recovery	Original D/H	Recovery R/S	Own duplicate	Own unwanted	Mean reads	Mean dispatches
Response retry	1024/3072	1280/3072	1024/3072	1024/3072	256/3072	0	17/12
Readback state	1792/3072	1792/3072	1792/3072	256/3072	256/3072	5/2	5/4
Separate reference	1792/3072	1792/3072	1792/3072	256/3072	256/3072	5/2	5/4
ConseqaX one final read	1792/3072	1792/3072	1792/3072	256/3072	256/3072	5/4	5/4
ConseqaX polling	1792/3072	1792/3072	1792/3072	256/3072	256/3072	5/2	5/4

TABLE III

STUDY A: ORIGINAL-CONTRACT FINAL REPORTS. CONDITIONAL ERROR IS WRONG/DEFINITE; CORRECT YIELD IS CORRECT/PLANNED. BOTH POSITIVE AND NEGATIVE CONCLUSIONS COUNT. COMPARISONS USE FULL-LEDGER TRUTH, RATHER THAN CONTEMPORANEOUS SUPPORT OR THE LATER RECOVERY TARGET.

Policy	State error	State correct yield	Linked error	Linked correct yield
Response retry	1024/1792	768/3072	768/1792	1024/3072
Readback state	0/2304	2304/3072	0/1024	1024/3072
Separate reference	0/2304	2304/3072	0/2048	2048/3072
ConseqaX one final read	0/2304	2304/3072	0/2048	2048/3072
ConseqaX polling	0/2304	2304/3072	0/2048	2048/3072

TABLE IV

STUDY B: EXACT FINITE-GRID REPORTS AND READ COSTS, 192 PLANNED/COMPLETED WORLDS PER POLICY. EVERY DEFINITE REPORT IS HINDSIGHT-CORRECT IN THIS UNIFORM-SUCCESS DESIGN. IT CANNOT ESTIMATE THE RESPONSE POLICY'S FALSE-SUCCESS RISK. LINKED ABSTENTION IS DISTINCT FROM STATE ABSTENTION.

Policy	State correct/planned	State unknown	Linked correct/planned	Total reads	Mean decision (ms)
Response retry	192/192	0/192	192/192	0	2
Readback state	81/192	111/192	0/192	741	224.375
Separate reference	81/192	111/192	81/192	741	224.375
ConseqaX one final read	3/192	189/192	3/192	192	150
ConseqaX polling	81/192	111/192	81/192	741	224.375

with no read, since all effects are constructed to succeed. Its report arrives before the effect and without horizon-covering evidence; this experiment supplies no estimate of its risk in failing worlds. Together, the studies show conditional polling utility and its cost, with no package superiority over the equally informed reference, no additional recovery-action benefit, and no task-population inference.

VI. SUPPORTING EVIDENCE AND VERSION BOUNDARIES

Earlier experiments corroborate narrower components without adding independent recovery families. Version 1.1.0 retained 503/504 factual-audit episodes across three small local models, with 72 unknown state reports; an 84-trace same-evidence judge made nine wrong among 60 definite state reports. Model-free tool transfer retained 656 cells and 900 operation captures, without autonomous reward. The native CAR bridge matched eligible point vectors in 640/1,024 paired groups; 384 were unavailable [17], [18]. Version 2.0 offline migration preserved all 503 retained canonical conclusions while changing 275 diagnosis codes. The historical manuscript and versioned evidence ledger retain denominators, protocol deviations, and detailed results.

The new experiment uses selected 2.1.1 public recording, contract, observation, and record-verification paths [7]. Its callback-based application polling and safety gate are explicit integration code. It does not evaluate the public blocking polling helper, composite/multiple-source verification, OpenTelemetry, SDK integrations, arbitrary multi-action traces, or the full release surface. Historical 1.1.0 results and 2.0 migration are not relabeled as 2.1.1 performance. Byte-identical numerical modules alone would not establish end-to-end version equivalence.

VII. DISCUSSION AND LIMITATIONS

An evidence protocol is useful only under meaningful source and operation semantics. A watermark covers history through a declared time; finite auditing through S cannot prove absence of all future effects. Provider deduplication or terminal resolution is supplied equally to every arm, not created by verification. Concurrent external actors remain a distinct source of harmful global count changes. These assumptions constrain any recovery conclusion.

Strong-reference parity should be interpreted directly. Separate arithmetic reduces reliance on one verifier implementation, but a shared controller author and application state machine can preserve shared mistakes. The saved-artifact auditor uses separate code, while its author also implemented the controllers; this is not external replication. Exact agreement on this suite is neither superiority nor a proof of general safety. Advantages over response or state ablations must identify the

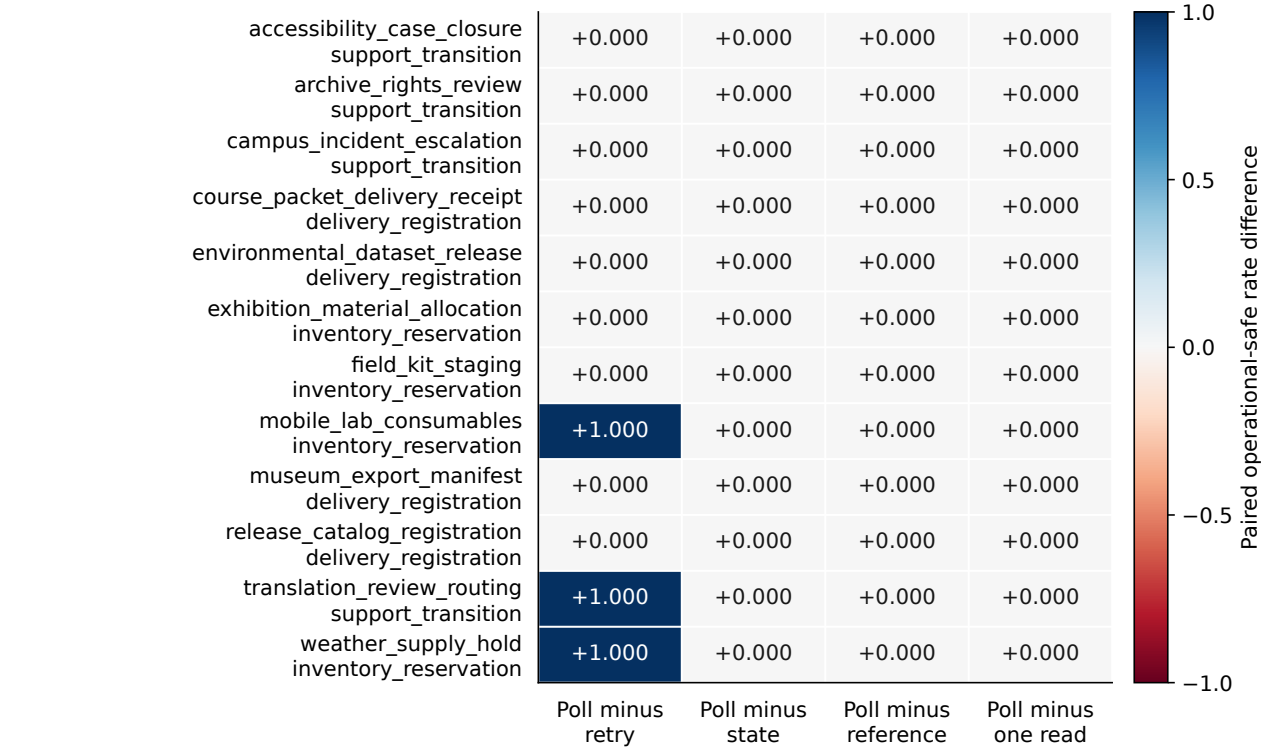


Fig. 2. Study A: every planned family's paired operational-safe difference, on a fixed $[-1, 1]$ scale. Each family has two related cases and 128 seeds; repetitions added no recorded-metric variation. The equal-family difference is +0.25 against response retry and zero against all three readback comparators. These purposive families do not support task-population inference.

omitted evidence and the precise claim or decision it changes. If provenance changes only reporting, that is the measured boundary; a causal action benefit cannot be inferred.

The task set is small, purposive, AI-authored, and implemented on one local substrate. Three record domains broaden application vocabulary and predicate use without establishing transfer to independent services. Repeated timing seeds cannot repair this external-validity gap. Deterministic policies exercise recovery logic, not language understanding, user negotiation, planning, provider authentication, or autonomous tool selection. These data can characterize a conditional decision boundary; they cannot establish general autonomous-agent improvement or effectiveness across independently operated services.

Evaluation uses executable application contracts, effect-ledger ground truth, and separately implemented automated checks. No participant study or annotation study is part of this paper; interpretability, preference, rationale quality, and user trust are not measured. Automated cross-checks can expose implementation or arithmetic inconsistencies, but cannot establish that an external adapter's source-admission declarations or provider guarantees are true. That requires objective validation against the service's actual semantics and independently retained effects.

The next external study should seal independently selected tasks in a maintained agent environment or provider sandbox,

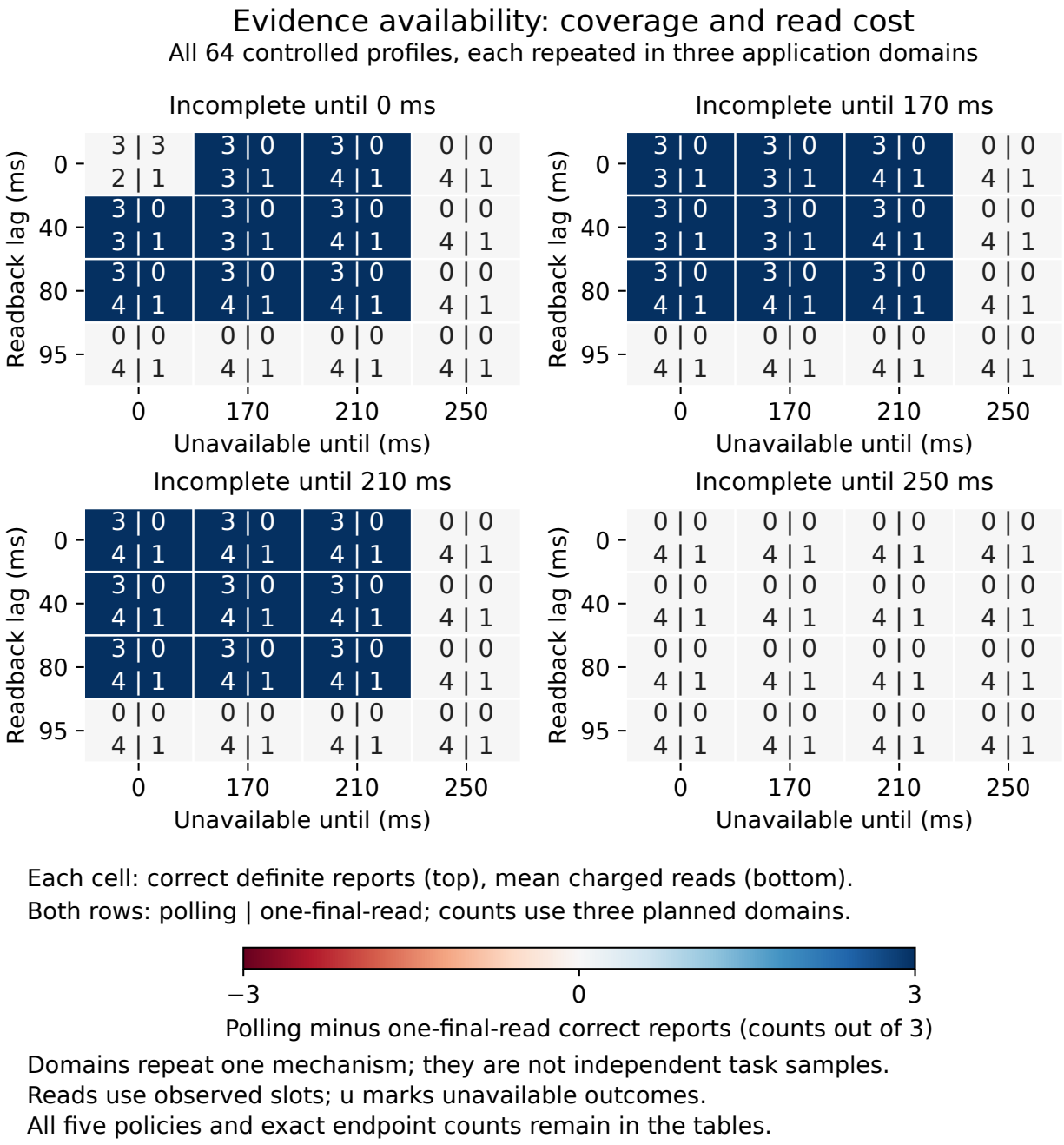


Fig. 3. Study B: all 64 controlled availability profiles, repeated across three domains. Actual report coverage and read costs are retained for polling and one-final-read. These paired conditions isolate evidence timing; every policy has identical successful physical outcomes, and the classical reference matches polling.

freeze a real agent policy, and compare the classical same-key recovery policy under identical capabilities and budgets. It must retain native reward plus an independently checked side-effect ledger and verify adapter/status/idempotency semantics. A declared task-sampling design and family-level precision target are necessary for broader inference. More internal cases, AI audits, or reference-action replay cannot substitute for this validation.

VIII. Conclusion and Artifact Availability

ConseqaX provides inspectable boundaries between tool responses, verified temporal state, operation-linked origin, and explicitly recorded assertions and evidence. The closed-loop study evaluates the application decisions and resource costs that follow from those boundaries under shared provider guarantees. Its scope is a frozen challenge suite and deterministic recovery policies. Historical model, transfer, replay, and migration evidence retains its original versions and targets. Comparisons use established provider guarantees as shared assumptions and treat reference agreement as conformance.

The current MIT distribution is version 2.1.2 [19]; these experiments retain their 2.1.1 pin [7]. The public repository retains protocols, source hashes, schedules, raw traces, dispositions, evaluator arithmetic, and deviations. A DOI-backed immutable research archive has not yet been established. The earlier manuscript preserves full replay definitions, interval tradeoffs, source pins, and historical audits. No real payment or production-service write is part of the reported recovery experiments.

AI Assistance and Verification Disclosure

Generative AI assisted software development, case creation, source inspection, automated artifact checking, and manuscript drafting. Separate AI authorship and saved-artifact audits do not constitute external replication. No participant or annotation study was conducted. The author is responsible for the manuscript and venue-required disclosure.

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